# Tide retrospective: the analytic-germ instantiation

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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## 1 Setting

Sixth tide of the germbij arc. The five merged tides carry their analytic input as the factored hypothesis $c\,w^m \le |a(w)|$ on $[0, r_0]$ (or its scaled-set analogue): honest, but one step removed from the note's actual hypothesis, real analyticity with nonzero germ. The germbij note says "analyticity enters only through finite vanishing order"; this tide makes that sentence a theorem, discharging the factored hypothesis from analyticity and thereby closing the gap between the Lean chain and the note's setting. It also begins the annotation phase: the note now carries formalisation markers (the primer's `leanref` convention, margin squares hyperlinked to the pinned source) on its Lemma 7.1, Lemma 7.2, and the proof of Theorem 7.3.

## 2 The thing formalised

One lemma, `analytic_growth_lower_bound` in `Laplace/Analytic.lean`, sorry-free: if $a$ is real analytic at 0 with nonzero germ (finite `analyticOrderAt`), then there exist $m$ (the order of vanishing), $c > 0$ and $r_0 > 0$ with

$$c\,w^m \;\le\; |a(w)| \qquad \text{on } [0, r_0].$$

The deliberation called this "the missing reusable bridge from finite analytic order to the already-merged quantitative hypothesis". Composed with the merged chain, the germbij identifiability bound now follows from analyticity of the potentials rather than from a growth hypothesis assumed outright; the explicit composition (the analytic identifiability corollary) is deferred to a follow-up per the vote.

## 3 Proof strategy

Mathlib's analytic-order API does the heavy lifting: `AnalyticAt.analyticOrderAt_eq_natCast` factors $a(w) = w^m\,g(w)$ eventually near 0, with $g$ analytic and $g(0) \neq 0$. Continuity of $|g|$ gives $|g| > |g(0)|/2$ on a neighborhood (`eventually_gt_nhds` through `ContinuousAt.abs`); conjoining the two eventual facts and extracting one radius (`Metric.eventually_nhds_iff`), then halving it to clear the strict-inequality ball boundary, gives the interval; on it, $|a(w)| = w^m |g(w)| \ge (|g(0)|/2)\,w^m$.

## 4 Roadblocks and resolutions

None: the file compiled on the first attempt with zero warnings, while the deliberation consult was still in flight. When the consult returned it endorsed the candidate and flagged exactly two proof subtleties (halve the extracted radius; normalise the factorisation with `sub_zero`, `smul_eq_mul`, `abs_mul`), both of which the implementation already handled. This is the arc's third first-attempt compile, evidence that the accumulated playbook (stage `nlinarith` hints, parenthesise nested integrals, `exact` before `simpa`) has converged on this corner of Mathlib.

## 5 What was Mathlib, what was new

Almost everything was Mathlib: the order and factorisation API,[1] and the continuity and filter idioms. New: only the statement's packaging, chosen so that its conclusion is verbatim the hypothesis of `sector_lower_bound`.

## 6 Lessons

When a paper phrase like "analyticity enters only through finite vanishing order" names a single mathematical mechanism, check whether Mathlib already owns that mechanism before planning any infrastructure: here the entire tide was one existing iff-lemma plus filter bookkeeping.

## 7 Follow-ups

- The composed corollary (candidate B): the 1D identifiability bound under analytic hypotheses outright, quantifiers arranged as constants first, then all sufficiently large $t$.
- The multivariate analogue: from a nonzero leading homogeneous part in $d$ variables, produce the scaled set $S$ with its amplitude bound (the cap of directions where the leading part is bounded away from zero).
- Continue the note annotation: bump the pinned commit and tag the instantiation once merged.

[1] `Mathlib.Analysis.Analytic.Order`.